a will usually be 1

c = $(\frac{b}{2})^2$

still works with odds

for example

$x^2 + 11x + c$
$c = (\frac{11}{2})^2$
$c = 5.5^2$
$c = 30.25$

EX:

$y = x^2 + 8x + 21$

y int = $(0, 21)$

POR = $(-8, 21)$

minus 21 from both sides

$y - 21 = x^2 + 8x$

add c (AKA $(\frac{b}{2})^2$) to both sides

$y - 5 = x^2 + 8x + 16$

use PS to factor rhs

$y = (x + 4)^2$
$y = (x + 4)^2 + 5$

Vert = $(-4, 5)$

EX:

$y = -3x^2 - 24x - 41$

y int = $(0, -41)$

POR = $(-4. - 41)$

$y + 41 = -3x^2 - 24x$

GCF is -3

$y - 7 = -3(x^2 + 8x + 16)$
$y - 7 = -3(x + 4)^2$
$y = -3(x + 4)^2 + 7$

Vert = (-4,7)